



Legal infrastructures for firm innovation: Effects of intellectual property protection in the era of digital economy

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ABSTRACT

This paper, using a sample of A-share firms in China from 2018 to 2022, demonstrates a positive influence of intellectual property protection on firm innovation and reveals that digital transformation mediates this relationship, amplifying the innovation impetus provided by robust intellectual property regimes. Notably, the positive effect of intellectual property protection on firm innovation is found to be weaker high-tech industries than that in non-high-tech industries. These findings highlight the nuanced role of intellectual property legal frameworks in fostering innovation and the evolving dynamics of digital transformation across different industry contexts.

1. Introduction

In the contemporary digital economy, technology and data are pivotal for shaping the competitive edge of firms. Intellectual property (IP) protection, a critical legal infrastructure, supports innovation (Neves et al., 2021). China, a pivotal player in global innovation, presents a unique approach to IP protection. As digital transformation becomes the norm in China, the intersection of IP rights and the digital economy emerges as a complex and dynamic field. IP protection serves as both a legal issue and a strategic tool for fostering innovation and maintaining competitive advantage. The Chinese model, balancing IP protection with technological progress and public interest, provides valuable insights for enhancing innovation capabilities in the digital economy.

Despite the extensive literature on the relationship between IP protection and firm innovation (Neves et al., 2021; Song et al., 2024), there are several gaps have not been addressed. On the one hand, most studies have concentrated on the direct relationship between IP protection and innovation without considering the mediating role of digital transformation. This is a significant oversight, as digital technologies are increasingly integral to innovation processes. On the other hand, research often fails to account for the specific contexts in which IP protection operates, such as the varying effectiveness of IP laws across different industries, especially in the digital economy.

To address these gaps, this study, grounded in the property rights theory, explores the nexus between intellectual property (IP) protection, digital transformation, and firm innovation. According to the theory, clear and enforceable IP rights incentivize innovation by securing returns on inventive investments (Kim and Mahoney, 2005). Our research extends this concept to show that IP rights stimulate innovation directly and indirectly through digital transformation, which aids in managing IP and reduces enforcement costs (Chen et al., 2023). High-tech industries, marked by rapid tech change and R&D intensity (Ortega-Argilés et al., 2010), are particularly responsive to IP protection and digital transformation, suggesting variations in the relationship across industries. Thus, the interaction between IP protection, digital transformation, and firm innovation is expected to be different in high-tech and other industries.

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Firms' innovative endeavors serve as a pivotal catalyst for economic expansion and societal advancement (Wong et al., 2005). Securing an advantageous position through innovation, particularly in the digital economy, emphasizes the importance of IP protection (Neves et al., 2021). Legal safeguards provided by IP protection secure innovative milestones and stimulate R&D investment (Fan et al., 2013). As companies undergo digital transformation, the prominence of IP protection increases, which underscores its indispensable role in the contemporary business landscape. Digital transformation involves the integration of advanced technologies and the reshaping of organizational frameworks, leading to the creation of substantial IP assets. The protection of these assets is crucial for successful digital transitions. Emerging technologies like AI, cloud computing, big data, and blockchain enhance innovation capabilities and market competitiveness. Despite existing research, the mediating role of digital transformation between IP protection and innovation remains understudied (Neves et al., 2021; Song et al., 2024).

While existing research reveals the impact of IP protection on firm innovation, few studies discuss differences across industries. The relationship between IP protection and firm innovation is influenced by factors such as industry characteristics, market environment, and policy support (Chen et al., 2023). For instance, Chinese government measures to combat IP infringement and counterfeit products promote innovation, especially in industries with higher IP risks. Thus, the high-tech industry, due to its reliance on innovation and technology intensity, may play a unique moderating role (Chen et al., 2023).

We contribute to the existing literature by investigating and validating the mediating role of digital transformation in the relationship between IP protection and firm innovation. It reveals how IP protection enhances firms' innovation through the promotion of digital transformation, thereby providing a new perspective on the mechanism by which intellectual property protection influences firm innovation. Moreover, the paper further expands the existing theoretical model by incorporating high-tech industries as a moderating variable, offering new insights into the mechanism of intellectual property protection in different industry contexts.

2. Theoretical analysis and research hypotheses

IP protection incentivizes innovation by granting temporary monopolies to inventors, spurring private investment in innovation (Song et al., 2024; Fan et al., 2013). It acts as a catalyst for firm innovation, as it can encourage R&D investments and secure proprietary information against imitation or theft (Li, 2017; Fan et al., 2013). Strong IP laws increase patent acquisition and citations (Chen et al., 2023), as evidenced by Apple's success with robust patent protection. By reducing negative externalities associated with R&D and curbing illegal copying, IP protection enhances expected returns on R&D investments, further promoting innovation within firms (Fan et al., 2013). Therefore, this paper proposes the following hypothesis:

H1. The intellectual property protection positively affects firm innovation.

IP protection is crucial for fostering digital transformation and enhancing firms' innovation capabilities by safeguarding technology rights and encouraging R&D investment (Wen and Deng, 2023). This leads to increased application of digital technologies and novel product creation, which propels digital transformation. Additionally, IP protection enlivens technology markets, facilitating access to cutting-edge technologies and accelerating digital transformation. Digital transformation, in turn, boosts innovation by integrating digital tools and enhancing big data analytics and organizational adaptability, which allows firms to quickly meet market demands and innovate (Li et al., 2023; Xu et al., 2024). For instance, Amazon has leveraged its IP to manage digital transformation, and thus drove innovation and growth. Therefore, this paper proposes that digital transformation mediates the positive impact of intellectual property protection on firm innovation, and the hypothesis is as follows:

H2. Firm digital transformation mediates the effect of intellectual property protection on firm innovation.

This study holds that high-tech industries' rapid technological turnover diminishes IP's positive influence on innovation, as competitors can swiftly circumvent protections or innovate anew. Conversely, non-high-tech sectors benefit from slower tech evolution, allowing IP to provide enduring competitive advantages. Furthermore, high-tech innovation's complexity and interdisciplinary nature can outstrip IP protection, potentially impeding necessary technological integration (Lin et al., 2015). High-tech firms' emphasis on open innovation and knowledge sharing (Suh et al., 2004) may view stringent IP as an obstacle to knowledge dissemination and collaborative innovation. Thus, traditional firms might leverage patents for a competitive edge, unlike tech start-ups in dynamic sectors where innovation is influenced by various factors. Therefore, this study proposes the following hypothesis:

H3. The effect of intellectual property protection on corporate innovation will be stronger in non-high-tech firms than that in high-tech firms.

3. Study design

3.1. Sample data

The data on intellectual property protection is collected from the official website of the State Intellectual Property Administration of China, and other data is from the CSMAR database. We select Chinese A-share listed firms from 2018 to 2022 as the initial sample, eliminating firms with missing values and those in the financial industry. We also exclude firms with ST, ST*, or PT status. Continuous variables are winsorized at a 1 % tail.

3.2. Selection of variables

Independent variables (Intellectual Property Protection, IPP): This study employs regional IP protection intensity as a metric for IP protection effectiveness. Given that various factors influencing IP protection are uniform across China, we use the natural logarithm of "number of IP cases settled in a city + 1" to gauge regional IP protection intensity (Lai et al., 2020). This proxy reflects the legal enforcement of IP laws in different regions. It is selected due to its direct correlation with the jurisdiction's legal framework's effectiveness in resolving IP disputes and its validation through its link with other IP strength indicators such as patent filings and enforcement actions.

Dependent variable (Firm Innovation, FI): We use the natural logarithm of "the total number of jointly applied patents in the year + 1" to measure the innovation level of the firm in the year.

Mediating variable (Firm Digital Transformation, FDT): Following Wu et al. (2021), we construct a firm digital transformation indicator system by extracting keywords using Python. Annual reports which summarize firms' past and future strategies, are crucial for assessing strategic adjustments and business philosophies. By counting keyword frequencies in these reports, we can evaluate a firm's digital transformation more objectively. We first collect and convert annual reports into text using Python's Pdfplumber library. Then, we tally "digital transformation" keywords (see Table 1 for examples) to derive initial indicators. Considering the right-skewed nature of this data, we apply the natural logarithm to the sum of word frequencies to obtain a comprehensive index of firm digital transformation (Wu et al., 2021).

Control variables: We include firm size, firm age, return on equity (ROE), cash flow ratio, the rate of growth in operating income, duality that refers to the chairman and the general manager are the same person, share proportion of the largest shareholder (SPLS). Table 2 lists the description of all variables (see Table 2).

3.3. Model construction

As per the initial hypothesis, IP Protection affects firm innovation via firm digital transformation. This study develops three models: one for the direct IP protection-innovation link, another for IP's impact on intermediaries, and a third for intermediaries' effect on innovation. Validation of Hypotheses 1–3 would confirm significant correlations for parameters β_1 , β_2 , and β_3 (Song et al., 2024).

$$FI = \alpha_1 + \beta_{1it}IPP + \gamma_{1it} Controls + \varepsilon_{it} \quad (1)$$

$$FDT = \alpha_2 + \beta_{2it}IPP + \gamma_{2it} Controls + \varepsilon_{it} \quad (2)$$

$$FI = \alpha_3 + \beta_{3it}IPP + \delta_{it}FDT + \gamma_{3it}Controls + \varepsilon_{it} \quad (3)$$

3.3.1. Baseline regression test model

As mentioned above, model (1) is constructed to test the positive relationship between intellectual property protection and firm innovation.

In model (1), FI is firm innovation; IPP is the intensify of intellectual property protection; Controls are the control variables; β and γ are the regression coefficient, α_0 is the constant term, i and t indicate firm and time respectively; and ε_i , ε_t are random disturbance terms.

3.3.2. Mediating effect test model

To test the mediating role of firm digital transformation on the relationship between IP protection and firm innovation, we construct the model (2) and model (3) as above.

Model (2) tests the impact of IP protection on firm transformation. FDT indicates firm digital transformation. Model (3) tests the impacts of firm digital transformation on firm innovation. And others have the same meaning as Model (1).

Table 1
Examples of digital transformation keywords.

Dimensions	Examples of keywords
Digital technology	Artificial intelligence, intelligent robot, natural language processing, biometrics, machine learning, deep learning, face recognition, autonomous driving, cloud computing, stream computing, multi-party secure computing, distributed computing, brain-like computing, EB level storage, Internet of Things, blockchain, digital currency, big data, text mining, data visualization, augmented reality, virtual reality, data mining, etc.
Application of digital technology	NFC payment, C2C, O2O, B2B, B2C, C2B, e-commerce, mobile payment, smart wear, smart transportation, intelligent customer service, smart medical care, smart culture and tourism, smart grid, digital marketing, digital finance, fintech, internet finance, etc.

Table 2
Variable description.

Variable	Variable symbols	Description
Intellectual Property Protection	IPP	Ln (the number of intellectual property cases settled in a city + 1)
Firm innovation	FI	Ln (R & D investment +1)
Firm digital transformation	FDT	Ln (the sum of word frequencies of the keywords of "digital transformation")
Firm innovation	FI	Ln (the total number of jointly applied patents in the year +1)
Firm size	Size	The natural logarithm of total assets
Return on equity	ROE	Net profit/average balance of shareholders' equity
Cash flow ratio	Cashflow	Net cash flow from operating activities divided by total assets
The rate of growth in operating income	Growth	Ration of the revenue of this year to last year's revenue minus 1
Duality	Dual	The chairman and the general manager are the same person = 1, otherwise = 0
Share proportion of the largest shareholder	SPLS	Number of shares held by the largest shareholder / total number of shares
Firm age	Age	Ln (number of years since listing+1)

4. Empirical analysis

4.1. Descriptive statistics

In order to get a preliminary understanding of the relationship between the descriptive variables, we conducted a descriptive analysis. of the variables. [Table 3](#) presents the descriptive statistics. IP protection has a mean of 7.503, SD of 2.064, and a range from 1.609 to 10.74, reflecting substantial variation. Firm innovation's mean is 0.916, median is 0, SD is 1.356, with a range from 0 to 5.557, indicating significant dispersion in innovation levels across the sample (see [Table 3](#)).

4.2. Correlation analysis

To preliminarily test the hypotheses, we conducted a correlation analysis. The correlation coefficients of variables in our study are listed in [Table 4](#). The correlation coefficient of IP protection with firm innovation is 0.059, which is positive and significant at the 1 % level (see [Table 4](#)), which indicates a positive relationship between IP protection and firm innovation.

4.3. Baseline regression results

In order to test H1 that IP protection can positively affect enterprise innovation, this paper uses OLS method for regression. The results for the baseline regression are presented in [Table 5](#). In column (1), firm innovation is used as explanatory variable while IP protection is the explanatory variable, and the results show that there is a positive relationship between IP protection and firm innovation, as the regression coefficient of IP protection are 0.0494, which is significant at the 1 % level (see [Table 5](#)). These results indicates that IP protection can significantly promote firm innovation, supporting H1.

4.4. Robustness checks

This study conducts additional robustness checks to further verify the findings. Firstly, we adopt an alternative measure of IP protection—the logarithm of "number of patents granted in a city +1" (PG) and reexamine the relationship between IP protection and firm innovation. Secondly, we also use the logarithm of "number of patents obtained of a firm +1" (FI2) to remeasure firm innovation and conduct the baseline regression again. The regression results are presented in [Table 6](#). As shown in [Table 6](#), PG positively affect FI at the 1 % significance level, and IPP has a positive effect on FI2 and the estimated coefficient of IPP is significant at 1 % level (see [Table 6](#)), which further provide empirical support for H1, indicating the robustness of our baseline regression.

Moreover, in order to solve the possible endogeneity problem between IP protection and firm innovation, this paper lags the explanatory variables—IP protection by one period and then adds them to the baseline regression, and the regression results are shown in [Table 7](#). According to [Table 7](#), the estimated coefficient of the explanatory variable lagged one period is 0.0511, and it is still significant at the level of 1 % (see [Table 7](#)), which further supports the results of the previous baseline regression.

Table 3
Descriptive statistics.

Variable	N	mean	p50	sd	min	max
IPP	17,688	7.503	7.643	2.064	1.609	10.74
FI	17,688	0.916	0	1.356	0	5.557
Size	17,688	22.31	22.09	1.330	19.97	26.43
ROE	17,688	0.053	0.073	0.159	-0.814	0.368
Growth	17,688	0.135	0.094	0.344	-0.588	1.850
Dual	17,688	0.323	0	0.468	0	1
SPLS	17,688	0.329	0.305	0.146	0.083	0.727
Age	17,688	2.047	2.197	0.989	0	3.401

Table 4
Correlation analysis results.

	1	2	3	4	5	6	7	8
IPP	1							
FI	0.059***	1						
Size	0.0120	0.456***	1					
ROE	-0.00700	0.076***	0.087***	1				
Growth	-0.00100	0.038***	0.051***	0.315***	1			
Dual	0.069***	-0.100***	-0.208***	0.025***	0.033***	1		
SPLS	0.016**	0.063***	0.178***	0.169***	0.00300	-0.056***	1	
Age	-0.094***	0.193***	0.472***	-0.181***	-0.104***	-0.273***	-0.064***	1

Note: t-statistics in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 5
Regression results.

Variables	(1) FI	(2) FDT	(3) FI
IPP	0.0494*** (11.3125)	0.0739*** (16.6362)	0.0446*** (10.1118)
Size	0.5207*** (57.3942)	0.1569*** (19.1774)	0.5104*** (55.6107)
ROE	0.1743*** (3.0725)	-0.1181* (-1.8555)	0.1820*** (3.2233)
Growth	-0.0634** (-2.4528)	-0.1115*** (-3.9628)	-0.0561** (-2.1707)
Dual	-0.0582*** (-3.1314)	0.1555*** (7.7179)	-0.0684*** (-3.6833)
SPLS	0.0650 (0.9760)	-0.6034*** (-9.5801)	0.1047 (1.5699)
Age	0.0516*** (5.0011)	-0.0536*** (-4.8650)	0.0552*** (5.3457)
FDT			0.0657*** (9.0413)
Constant	-11.3554*** (-55.5554)	-2.5673*** (-13.8346)	-11.1868*** (-54.2976)
Observations	17,688	17,688	17,688
R-squared	0.3011	0.3420	0.3043
Year	Yes	Yes	Yes
Industry	Yes	Yes	Yes

Note: t-statistics in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

4.5. Heterogeneity analysis

4.5.1. Heterogeneity analysis of industry

To examine the asymmetric effect of IP protection at the industry level, this study conducts a heterogeneity test by dividing the samples into high-tech and non-high-tech firms. The regression model tests H3, proposing that the effect of IP protection on firm innovation is weaker in high-tech firms than in non-high-tech firms. Table 8 presents the results of the heterogeneity analysis. As shown in Table 8, the effect of IP protection on firm innovation is positive and significant at the 1 % level in both high-tech (0.0416) and non-high-tech firms (0.0505) (see Table 8). The regression coefficient of IP protection is larger in non-high-tech firms, indicating a stronger impact on firm innovation compared to high-tech firms. Therefore, H3 is supported.

4.5.2. Heterogeneity analysis of region

We also conducted a heterogeneity test, segmenting samples by region—eastern versus central and western firms—and reassessed the IP protection-innovation link. Regression results in Table 9 show that IP protection significantly boosts innovation more in eastern (0.0492) than central and western firms (0.366) at the 1 % level (see Table 9). This regional disparity may stem from a more comprehensive innovation ecosystem in the east, with concentrated R&D, a rich talent pool, a vibrant market economy, and favorable policies that amplify the motivational effect of IP protection on innovation activities.

4.6. Analysis of mediation effects

To test the mediating role of firm digital transformation on the relationship between IP protection and firm innovation, we construct a stepwise mediation model, and the results are shown in Table 5. In column (2), the regression coefficient of IP protection on firm digital transformation is positive (0.0739) and significant at the 1 % level, indicating that IP protection improves firm digital

Table 6
Results of robustness checks of alternative measures of IPP and FI.

Variables	(1) FI	(2) FI2
IPP	0.0763*** (11.8606)	0.9955*** (10.2398)
PG	0.5215*** (57.5173)	
Size	0.1680*** (2.9659)	8.4541*** (26.8106)
ROE	-0.0633** (-2.4473)	0.1043 (0.1060)
Growth	-0.0617*** (-3.3171)	-2.7375*** (-6.0272)
Dual	0.0596 (0.8935)	0.1841 (0.5049)
SPLS	0.0552*** (5.3339)	7.4919*** (4.7990)
Age		-0.1634 (-0.8639)
Constant	-11.7626*** (-56.0486)	-193.9781*** (-27.2606)
Observations	17,688	17,688
R-squared	0.3015	0.1963
Year	Yes	Yes
Industry	Yes	Yes

Note: t-statistics in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 7
Results of robustness checks of IPP lagged one period.

Variables	(1) FI
IPP	0.0511*** (9.7434)
ROE	0.2111*** (3.2888)
Growth	-0.0849*** (-2.8187)
Dual	-0.0632*** (-2.8541)
SPLS	0.0301 (0.3824)
Age	0.0414*** (2.7880)
Constant	-11.8523*** (-51.3471)
Observations	13,145
R-squared	0.3130
Year	Yes
Industry	Yes

Note: t-statistics in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

transformation. In column (3), firm innovation is the dependent variable, and we include IP protection and digital transformation simultaneously. The regression coefficients of IP protection (0.0446) and digital transformation (0.0657) are both positive and significant at the 1 % level, but the coefficient of IP protection is smaller than in column (1) (0.0494). These results indicate that IP protection improves firm innovation by facilitating firm digital transformation, supporting H2, which proposes a mediating role of digital transformation in the relationship between IP protection and firm innovation.

5. Conclusion

This study investigates the relationship between IP protection and firm innovation, as well as the mediating role of firm digital transformation and the moderating effect of high-tech industry. The results show that: (1) IP protection promotes firm innovation, which provides empirical evidence for its positive impact in the digital economy era, despite some debate on its impact (Song et al., 2024); (2) Firm digital transformation mediates the relationship between IP protection and firm innovation, highlighting its

Table 8
Results of heterogeneity analysis of industry.

Variables	(1)	(2)
	FI High-tech industry	FI Non-high-tech industry
IPP	0.0416*** (6.0096)	0.0505*** (8.9303)
Size	0.5475*** (50.1342)	0.4534*** (26.9633)
ROE	0.3790*** (4.3087)	0.0768 (1.0368)
Growth	-0.1113*** (-2.6786)	-0.0386 (-1.1533)
Dual	-0.1074*** (-4.0768)	-0.0098 (-0.3733)
Top1	-0.2228** (-2.1609)	0.2958*** (3.3742)
ListAge	0.1245*** (5.9790)	0.0321** (2.5452)
Constant	-10.0488*** (-26.4046)	-11.7217*** (-47.5805)
Observations	8081	9607
R-squared	0.2130	0.3733
Year	Yes	Yes
Industry	Yes	Yes

Note: t-statistics in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 9
Results of heterogeneity analysis of region.

Variables	(1)	(2)
	FI Eastern region	FI Central and western region
IPP	0.0492*** (8.6483)	0.0366*** (4.2601)
Size	0.5363*** (50.3397)	0.4704*** (27.9113)
ROE	0.2175*** (3.1994)	0.1038 (1.0012)
Growth	-0.0582* (-1.8583)	-0.0664 (-1.4561)
Dual	-0.0633*** (-2.9370)	-0.0416 (-1.1467)
Top1	0.0033 (0.0416)	0.2268* (1.8225)
ListAge	0.0639*** (5.1105)	0.0220 (1.1927)
Constant	-11.6571*** (-47.6998)	-10.2304*** (-27.4457)
Observations	12,988	4700
R-squared	0.3136	0.2814
Year	Yes	Yes
Industry	Yes	Yes

Note: t-statistics in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

significance in leveraging IP protection to enhance innovation, a factor often overlooked in existing literature; (3) In high-tech industries, the effect of IP protection on firm innovation is weaker, which diverges from the widely held belief (Grzegorzczuk, 2020; Liu et al., 2020) and offers new insights into the role of IP protection.

Our study offers insights for market participants, regulators, and stakeholders regarding the nexus between digital transformation and IP protection in fostering innovation. Market participants should invest in digital technologies to amplify the benefits of IP protection, thereby enhancing innovation. Businesses in non-high-tech sectors should maximize IP protection, while high-tech firms should seek additional mechanisms to complement IP protection. Regulators must develop policies that encourage digital transformation and robust IP protection, including promoting digital literacy and establishing digital IP frameworks. Policymakers should extend IP support to industries beyond high-tech, as our findings indicate that IP protection positively influences innovation across various sectors. Investors should prioritize funding companies with strong IP strategies and ongoing digital transformation efforts.

This study has some limitations. The use of case settlements as a proxy for IP protection may not capture all dimensions. Unobserved factors might also influence our results. Future research should refine and validate these measures and explore alternative

proxies for IP protection. Additionally, this study's findings are based on data from Chinese firms, limiting generalizability. Future research should extend the analysis to multiple countries to better understand how IP protection influences innovation in the digital economy.

CRedit authorship contribution statement

Bingru Yuan: Writing – review & editing, Writing – original draft, Validation, Resources, Methodology, Conceptualization. **Weibo Li:** Writing – original draft, Methodology, Data curation.

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Data availability

Data will be made available on request.

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